INJECTIVITY OF MAPS OF SIMPLICIAL SETS TESTED ON NONDEGENERATE SIMPLICES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the componentwise step is the universal property of the coproduct; hypothesis (a) guarantees the index sets of the two decompositions agree.

Editorial note. The label lemma-injective-map-simplicial-sets names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Splitting of a simplicial set. Same unit, Lemma lemma-splitting-simplicial-sets. The proof decomposes U_n and V_n by the splitting of Lemma lemma-splitting-simplicial-sets.

Source reference. Original reference 017R: sources/stacks-simplicial/pinned-original-source.tex, source lines 2013–2013. Prerequisite pointer invoked at proof line(s) 2055; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : U \to V$ be a morphism of simplicial sets. Suppose that (a) the image of every nondegenerate simplex of U is a nondegenerate simplex of V and (b) the restriction of f to a map from the set of nondegenerate simplices of U to the set of nondegenerate simplices of V is injective. Then f_n is injective for all n. Same holds with "injective" replaced by "surjective" or "bijective".*

Proof. Under hypothesis (a) we see that the map f preserves the disjoint union decompositions of the splitting of Lemma 017R, in other words that we get commutative diagrams

$$\begin{array}{ccc} \coprod_{\varphi:[n]\to[m] \text{ surjective}} N(U_m) & \longrightarrow & U_n \\ \downarrow & & \downarrow \\ \coprod_{\varphi:[n]\to[m] \text{ surjective}} N(V_m) & \longrightarrow & V_n. \end{array}$$

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And then (b) clearly shows that the left vertical arrow is injective (resp. surjective, resp. bijective). □